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SUPERSEDING
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FEDERAL SPECIFICATION

PIPE AND FITTINGS, PLASTIC, RIGID (STYRENE RUBBER, DRAIN)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers styrene-rubber plastic pipe, socket fittings and threaded adapters which are resistant to the action of organic matter, alkalies and acids normally existing in ground and surface water drainage systems.

1.2 Classification. Pipe and fittings covered by this specification shall be of the following length and sizes as specified (see 6.2):

1.2.1 Pipe.

Pipe size - 2, 3, 4, 5 and 6 inches
5.1, 7.6, 10, 13 and 15 cm
Length - 10 feet (3.1 m)

1.2.2 Fittings.

Couplings	All sizes
Reducers	To one size less
Wyes 45°	All sizes
Tees	All sizes
Saddle Tees	6 inches (15 cm)
Saddle Wyes	6 inches (15 cm)
Double Tees	6 inches (15 cm)
Double Wyes	6 inches (15 cm)
1/4 Bends	All sizes
1/8 Bends	All sizes
Crosses	2 through 6 inches (5.1 - 15 cm)
45 deg long sweeps	2 through 6 inches (5.1 - 15 cm)
90 deg long sweeps	2 through 6 inches (5.1 - 15 cm)
Adapters to cast iron	2 through 6 inches (5.1 - 15 cm)
Adapters to clay bell	2 through 6 inches (5.1 - 15 cm)
Adapters to asbestos cement	4 and 6 inches (10 and 15 cm)
Adapters to iron pipe thread	2, 3, and 4 inches (5.1, 7.6, and 10 cm)
Threaded cleanout adapters	2, 3, and 4 inches (5.1, 7.6, and 10 cm)
Threaded plugs	2, 3, and 4 inches (5.1, 7.6, and 10 cm)
P traps	2, 3, and 4 inches (5.1, 7.6, and 10 cm)
Running traps	2, 3, and 4 inches (5.1, 7.6, and 10 cm)
Running traps with vent	2, 3, and 4 inches (5.1, 7.6, and 10 cm)
Caps	All sizes

2 APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

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Federal Specifications:

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Box, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-650 - Crates, Wood, Open and Covered.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

D2852 - Specification for Styrene-Rubber Plastic Drain and Building Sewer Pipe and Fittings.
D3122 - Specification for Solvent Cements for Styrene-Rubber Plastic Pipe and Fittings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P St., N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Pipe and fittings. Plastic pipe and fittings shall meet the requirements of ASTM D2852 and the following (see 4.3):

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(a) Styrene-rubber pipe and fittings may include, in addition to clean rework material, clean rejected or damaged pipe that has not been in service.

(b) Solvent cement. When specified (see 6.2), solvent cement conforming to the requirements of ASTM D3122 shall be provided for joint connections (see 4.3.2).

3.2 Pipe. Pipe shall be furnished with both ends plain or with one end plain and the opposite end belled, as specified (see 6.2). When specified (see 6.2), pipe with plain ends shall have a coupling bonded to one end.

3.3 Length of pipe. When specified (see 6.2), pipe may be furnished in lengths other than 10 feet (3.1 m).

3.4 Workmanship. Pipe and fittings shall be homogeneous, free from visible cracks, holes, and foreign inclusions. Pipe shall be uniform in color opacity, and density.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of each size and each length of pipe, or each size and each style of fitting, offered for delivery at the same time.

4.2.2 Visual, dimensional and workmanship. Pipe and fittings shall be examined for these characteristics in accordance with table I and MIL-STD-105, using inspection level S-3, and an Acceptable Quality Level (AQL) of 2.5 percent defective per each 100 units for major defects, and 4.0 percent defective per each 100 units, for minor defects.

TABLE I. Classification of defects, end item

Defects	Major	Minor
Size of pipe not as specified.	X	
Type and size of fittings not as specified.	X	
Solvent cement for joint connections not supplied when specified.		X
Plain end pipe not supplied with couplings when required.		X
Pipe length and diameter not as specified.	X	
Pipe and fitting dimensions and tolerances not in accordance with ASTM D2852.	X	
Cracks in pipe or fittings.	X	
Holes in pipe or fittings.	X	
Damage or defects, other than those listed, affecting serviceability.	X	
Damage or defects, other than those listed, not affecting serviceability.		X

4.2.2.1 Pipe dimensions. Measure inside and outside diameters, wall thickness, and length in accordance with ASTM D2852.

4.2.2.2 Fittings. Measure wall thickness and socket dimensions in accordance with ASTM D2852.

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4.2.2.3 Inspection of preparation for delivery. An inspection shall be made to determine that packaging, packing and marking requirements comply with section 5 of this specification. Defects shall be scored in accordance with Table II. For examination of contents the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II and the AQL shall be 4.0 defects per hundred units.

TABLE II. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

4.3 Test methods.

4.3.1 Pipe and fittings. Test pipe and fittings in accordance with ASTM D2852 (see 3.1).

4.3.2 Solvent cement. Test solvent cement in accordance with ASTM D3122 (see 3.1).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging.

5.1.1.1.1 Fittings. Fittings of like description, in quantitite as specified (see 6.2) shall be packaged together in a close-fitting box conforming to PPP-B-636, class weather resistant. The box shall be closed in accordance with the appendix to the box specification.

5.1.2 Level B (civil agencies).

5.1.2.1 Unit packaging.

5.1.2.1.1 Fittings. Fittings of like description, in quantities as specified (see 6.2) shall be packaged together in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.1.3 Level C. The pipe and fittings shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Fittings. Fittings of like description shall be packed together in close-fitting boxes conforming to PPP-B-585, class 3, PPP-B-591, class II, PPP-B-601, type II, class 2, PPP-B-621, class 2, PPP-B-636, class weather resistant or PPP-B-640, class 2, grade A. The boxes shall be closed in accordance with the requirements to the applicable specification or appendix thereto. The gross weight of the triple wall fiberboard box and the wood boxes shall not exceed 200 pounds. The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the bos specification. Strapping shall be in accordance with the appendix to the applicable box specification.

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5.2.1.2 Pipe. Pipe of like description shall be packed together in a close fitting box conforming to PPP-B-640, class 2, or in a crate conforming to PPP-C-650, style A, type II. The box and crate shall be closed in accordance with the appendix to the applicable specification.

5.2.2 Level B.

5.2.2.1 Fittings. Fittings of like description shall be packed together in close-fitting boxes conforming to PPP-B-585, class 1, PPP-B-591, class 1, PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636, class domestic. The fiberboard box shall be closed in accordance with the appendix to the box specification. The gross weight of the wood boxes shall not exceed 200 pounds and the fiberboard boxes shall not exceed the weight limitation of the box specification.

5.2.2.2 Pipe. Pipe of like description shall be packed together in close-fitting boxes conforming to PPP-B-640, Class, or in a crate conforming to PPP-C-650. The boxes and crate shall be closed in accordance with the appendix to the applicable specification.

5.2.3 Level C. Pipe and fittings shall be packed to insure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special markings required by the containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special markings required by the contract or order, all interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Styrene-rubber plastic pipe and fittings are intended for use in non-pressure drainage and certain other liquid wastes for the following applications:

(a) House connections to septic tanks.
(b) Footing drains (foundation drains).
(c) Storm drainage.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Pipe size and length (see 1.2.1 and 3.3).
(c) Fittings, type and size (see 1.2.2).
(d) Solvent cement (see 3.1).
(e) Pipe ends (see 3.2).
(f) Size of lot, if different from 4.2.1.
(g) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).

6.3 Pipe size. Foundation footing drains and storm drains should be 3 inch (7.6 cm) minimum pipe, but not less than the minimum size required for the installation served.

6.4 Pipe ends. Pipe covered by this specification is not threaded because of its relatively thin wall thickness. When threaded connections are required, use transitition adapter fittings.

6.5 Solvent cement. Solvent cement will provide sufficient open time for making good joints and connections but joints should be completed immediately after applying solvent cement. Should any delay develop in assembly, an additional coat of solvent cement should be applied immediately prior to joining. Follow cement manufacturer's recommendations on the container label for set time and cure time. Consult cement manufacturers for this data, and for detailed instructions, when necessary.

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6.6 Styrene rubber plastics. Plastics consisting of at least 50 percent styrene plastics combined with rubbers and other compounding ingredients.

6.7 Metric equivalents. The metric equivalents to dimensional requirements are for information only and are not intended to be used for determining acceptance or rejection of items otherwise conforming to the requirements of this specification.

MILITARY INTERESTS:

User activity:

DCSC-CS

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activities:

COMMERCE-NBS
HEW-FEC
INTERIOR-BPA

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